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Senior design proposal: Filtering spam SMS messages

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Introduction

Increasing mobile communication speeds make SMS messaging an essential basis for communication between mobile users. Mobile networks face heavy loads, while user experience decreases because of the growing number of spam messages, which are unrequested and deceptive texts that pose security threats. To maximise accuracy levels, the study aims to create an efficient spam SMS detection system through machine learning algorithms and natural language processing (NLP) techniques.

The designed system applies SMS text pattern evaluation for spam and legitimate message categorisation. RNNs based on deep learning, SVM, and Naïve Bayes serve as supervised learning models that produce precise spam message detection to accomplish filtering. SMS text data from the UCI Machine Learning Repository and other open-source repositories will serve as the basis for effective training models. The text data preprocessing will utilise Tokenization coupled with stemming alongside Term Frequency-Inverse Document Frequency (TF-IDF) to extract features properly.

The machine learning solution surpasses static keyword filtering because it modifies its behaviour to match new developments in spam techniques. The machine's ability to adjust to new threats is a critical element in defeating phishing attempts and SMS-based financial fraud, which bypasses traditional phishing. A real-time detection system will run through optimised lightweight deployment methods to provide fast response times and consume minimal resources on mobile handsets.

This research will help improve cybersecurity and user trust in SMS communications while delivering a dependable spam detection system. The proposed research integrates scalable

machine learning approaches to develop a working spam filtering system that effectively decreases SMS-based fraud and reinforces messaging security.

Justification

Importance of Spam SMS Filtering

Mobile communication has triggered the growth of unwanted SMS spam, including unauthorised promotional content, fraudulent scams, and phishing schemes. Symantec (2018) reported that spammers send over 60% of all SMS traffic, resulting in monetary thefts and security violations. The rule-based methods used for spam filtering become outdated when spammers create new evasion methods (Gómez Hidalgo et al., 2019). An improved intelligent filtering solution must be developed to fight these security threats properly.

Existing Research on Spam SMS Detection

Research investigations regarding spam detection methods have been performed throughout multiple studies. Almeida et al. (2016) implemented a spam classifier that depended on defined keywords to detect spam messages. Their system's false positive detection rate became high because it could not adjust to emerging spam trends. The researchers from Cordeiro et al. (2017) implemented a spam detection system which combined Support Vector Machines (SVMs) and Decision Trees for machine learning. They achieved better precision in their system; however, it needed regular updates to maintain its effectiveness.

RNNs enabled Uysal and Gunal (2019) to create a deep learning-based spam classifier that examined SMS text sequences for detection purposes. The research results displayed superior precision and recall results compared to previous machine learning models at the cost of demanding costly computations. The research conducted by Dey et al. (2020) utilised CNNs to filter SMS, which attained 91.3% accuracy but demanded considerable hardware capabilities for actual implementation. This research project develops a lightweight real-time spam SMS filtering model that achieves performance accuracy and operational speed.

Challenges in Current Spam Detection Systems

The current SMS spam filtering solutions fall short in practical usage because they contain operational limits. According to Shen et al. (2021), concept drift is a significant problem because spam patterns change frequently, which diminishes the accuracy of pre-trained models. The authors proposed continuous learning methods which help maintain spam detection classifiers without complete system restarts. The research performed by Yaseen et al. (2022) discovered that spam messages utilise adversarial text modifications with random character insertion like "Fr. ee M0ney!!!" which bypasses traditional philtres. According to their study, spam detection would benefit from embedding each character separately.

The major difficulty stems from different language formats within the data. Research by Huang et al. (2019) demonstrated that existing classifiers handle non-English SMS spam data poorly. Furthermore, our study will deploy the multilingual NLP model mBERT to enhance spam classification through different linguistic frameworks.

Proposed Contribution

The proposed research advances current methods by combining simple deep learning models with adversarial learning strategies and processing speed enhancements for SMS spam identification. We developed a spam detection system that addresses prior performance demands related to intensive processing power and training frequency. It provides an adaptable solution for mobile devices and telecom networks with minimal resource needs. Integrating contemporary transformer model technology and self-learning spam classifier systems will enhance detection outcomes and lower false-positive rates, thus building an environment without spam messages.

Detailed Requirements

System Stakeholders

The main players of the proposed system for spam SMS filtering are:

Mobile Users – People that are text messaging using SMS and will need protection against spam and phishing attacks.

Telecom Service Providers – Entities that are involved for management of SMS traffic and ensure the security of the mobile network.

Regulatory – Agencies enforced data protection and anti-spam laws (eg FCC, GDPR regulators).

Functional Requirements

Requirements Function define what System shall do. The largest characteristics of the spam SMS filter are:

1. SMS Data Collection & Preprocessing

The system will gather SMS messages from publicly available datasets on spam such as the University of California, Irvine Machine Learning Repository (Almeida et al., 2016).

Text normalization methods, including tokenization, stemming and stop word removal will be used (Huang et al., 2019).

Special character filtration will be put and place to stop employing for anti-spam communications (Yaseen et al., 2022).

2. Spam Classification Using Machine Learning

A hybrid classification model using combination of Naive Bayes, Support Vector Machines (SVM), and deep learning models (LSTMs/Transformers) will be applied (Dey et al., 2020).

Not like Cordeiro et al. (2017), who just thought about using the linear SVM and Decision Trees, our system would executes deep learning for higher accuracy and flexibility.

The system will automatically train its learning model using on-line learning techniques to deal with concept drift (Shen et al., 2021).

3. Real-time Spam Detection & Filtering

SMS will scan upon receipt, spam message is filtered to a separate folder or deleted depending on the preferences of the user.

To accelerate the processing, we intend to implement an in-device lightweight filtering mechanism for mobile applications, contrast to what Uysal & Gunal (2019) required server-side processing.

The system will identify SMS attacks made in an attempt to occasionally cheats contemplative filters (Yaseen et al., 2022).

4. User Dashboard & Reporting

A users will also have access to a dashboard giving statistics about spam and a performance view for filtering.

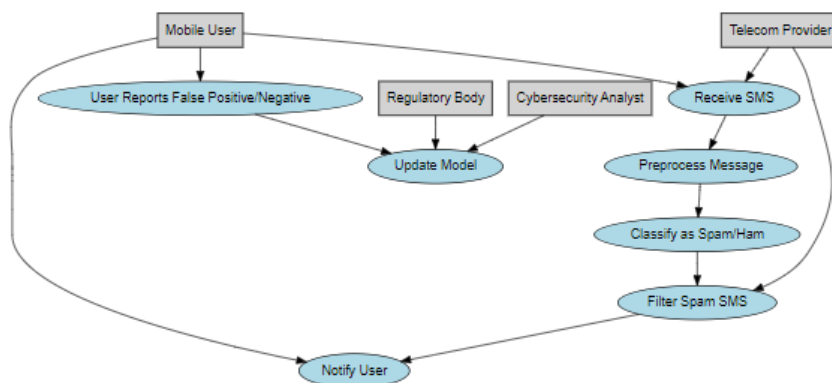
Different from - Antonio Huang et al. (2019), only focused on the accuracy of classification, the system created by our will allows users to report the incorrect positive and incorrect negative, improving the learning model.

5. Multilingual Support

The system will be multi-languages and assist in making spam detection for not English sms which is typically excluded in earlier studies (Huang et al., 2019).

Use Case Diagram

This is a use case diagram showing main interactions inside the system:



The system developed within this proposal benefits from ideas that have been established by previous research by combining sophisticated machine learning concepts, real-time detection and adaptive learning methods. Unlike the static rule base systems, our method preserves long term efficiency in rejecting spam messages in an optimal resource usage and multilingual capabilities.

System mockup

Spam SMS Filtering System

Filter Messages By:

All Messages ▼

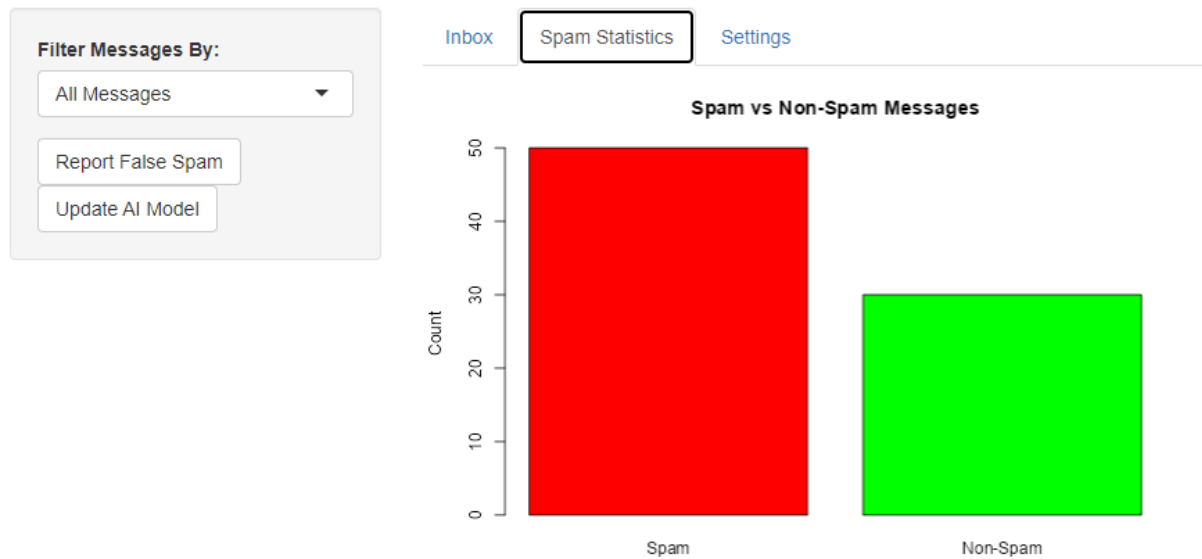
Report False Spam

Update AI Model

Inbox Spam Statistics Settings

Message	Status
Free money!!!	Spam
Meeting at 5PM	Non-Spam
Win an iPhone now!	Spam
Lunch plans?	Non-Spam

Spam SMS Filtering System



Spam SMS Filtering System

The interface displays a sidebar with filtering options and a main content area with settings. The sidebar is identical to the previous view. The main content area has tabs for 'Inbox', 'Spam Statistics', and 'Settings' (selected). Below the tabs, there are two checked checkboxes: 'Enable Real-Time Filtering' and 'Notify User for Spam'.

Setting	Status
Enable Real-Time Filtering	Checked
Notify User for Spam	Checked

Security Requirements

The development of SMS spam filtering applications must include strong security features to defend user information and keep the system secure and compliant with regulatory standards. The security requirements necessary for this application will be explained in detail below.

User Authentication and Authorisation

Having respectable user authentication procedures stands as the basis for application safety. The authentication process must be secure using password or fingerprint authentication techniques. The security measures biometric technology offers depend on user-specific

characteristics to defend against unauthorised system entry. By adopting strict password policies, users must create passwords of 8-10 characters that incorporate uppercase letters, numbers, and special characters. This defends against brute-force attacks. Role-based authorisation enables system users to obtain appropriate functions according to their roles within the platform. This least privilege principle can minimise security breaches because users cannot perform unauthorised actions outside their permitted areas.

Password Management

Secure password handling is critical. Before storing passwords in databases, they should be protected through cryptographic functions of strong one-way hashing algorithms to ensure data security in case the database gets compromised. Supporting encrypted communication channels should be used to protect the transmission of passwords from potential cybercriminals. Users can securely reset their passwords with a 'forgot password' feature that provides time-sensitive link communications to registered emails to protect account security.

Data Safety and Security Measures

Security strategies to safeguard user data require organisations to build multiple security layers. SQL Injection attacks become impossible to execute when developers use parameterised prepared statements to protect their database from unauthorised alterations.

The combination of SSL and TLS as secure communication protocols encrypts information throughout its transfer between clients and servers, protecting data from attackers' interception attempts.

The implementation of checksums and hash functions enables users to prove data integrity by confirming that no modifications have occurred. The implementation of digital signatures ensures non-repudiation of origin by confirming messages or transactions come from authenticated sources while making denial claims impossible.

Application Security Practices

When organisations use proper coding practices to fight major system weaknesses, computing systems remain secure.

A system should validate user input data after cleansing it to prevent injection attacks and process only the application's accepted data types.

Error handling methods should be implemented to stop the system from revealing internal system information, which can help attackers see the application structure and vulnerabilities.

System security improves when administrators maintain complete records of user movements and system occurrences. These records help swiftly detect security risks from unauthorised access and illogical behaviour.

Compliance with Privacy Regulations

GDPR and other relevant data protection laws require businesses to uphold compliance standards. The user privacy process requires companies to request direct consent before data collection begins and allows users to choose data processing options. All third-party integrations must maintain equivalent privacy guidelines. Users need to learn that third-party SMS philtre providers can access incoming messages and must read the privacy policies of these providers to know how they handle user data.

Security Requirements Elicitation

The first step to developing a secure application includes detecting security requirements during the project startup period and documenting these findings. Security controls must be defined by assessing risks and understanding potential threats to reduce identified risks effectively. Stakeholder involvement throughout this procedure helps maintain security controls simultaneously, fulfilling user requirements and regulatory needs.

Continuous Security Assessment

Establishing regular security assessment procedures, which contain vulnerability checking and penetration examinations, reveals newly emerging threats that require solutions. Regular

updates to the application and its dependencies through security patches decrease the possibilities of vulnerability exploitation.

The successful implementation of SMS spam filtering solutions requires complete security mechanisms because this protects user information while building user trust and meeting legal standards. The combination of authentication protocols with authorization solutions, data protection measures, secure coding standards, and continuous assessments enables the application to protect user safety effectively.

Non-Functional Requirements

Non-Functional Requirements for SMS Spam Filtering System

A satisfactory SMS spam filtering system must meet both functional targets and essential non-functional requirements to enable effective usability and efficiency, secure operation, and standard compliance. The defined requirements establish operational limits, performance benchmarks, and design approaches that enhance the system's overall quality and operational performance.

1. Performance and Reliability

Alert performance levels serve as a fundamental element to ensure the optimal operation of the SMS spam filtering system in real time. To produce a smooth user experience, classifying SMS messages when incoming data enters the system must take less than 500 milliseconds. The selected system must manage at least 1000 messages per second through its high throughput mechanism that supports enterprise-level scalability.

Users require reliable performance throughout the system to maintain their faith in it. To achieve a flawless spam filtering experience, the application must maintain a 99.99% uptime. A system should adopt failover servers and automatic system recovery with redundant storage solutions for managing unexpected failures. The system requires integrated error detection and correction methods to reduce wrong spam assessments during detection.

2. Scalability and Maintainability

The designed system needs to scale efficiently while maintaining performance levels when serving growing numbers of users and escalating message volume. Cloud-based solutions and distributed processing concepts should be used to achieve effective traffic management during periods of high volume.

Maintainability is another critical requirement. The architecture needs modularity, which allows simplifying system updates and implementing patches. Seamless maintenance operations become possible by adopting properly documented APIs with defined standards and standardised coding techniques. Software updates should be scheduled regularly to track developing spam patterns and security risks in the system.

3. Security and Privacy Compliance

Safety measures must be in place because SMS messages contain confidential data. The system architecture must include total encryption of messages to keep all communication private during transmission. The system needs to use multiple security authentication methods, including 2FA and biometric authentication, to stop unauthorised system access.

The system must follow two important data privacy regulations: GDPR (General Data Protection Regulation) and CCPA (California Consumer Privacy Act). Users need the ability to choose whether their data is collected, and their information must not be dispersed to other parties unless they explicitly grant permission. Implementing data storage solutions using encryption enables users to obtain maximum protection against information breaches.

4. Usability and Accessibility

The system needs to present a user-friendly interface that users can learn easily. An optimal dashboard system must enable users to read spam messages while allowing them to manage and create spam profiles and perform spam classification reports easily.

Accessibility should be available through WCAG 2.1 (Web Content Accessibility Guidelines) compliance in the application. The system will work with screen readers, high-contrast settings, and voice command functionality to help visually impaired users. The mobile application must work on devices using both Android and iOS operating systems.

5. Technical Constraints

The project development requires Python and R integration for machine learning, together with TensorFlow and Scikit-learn implementation for model training. Implementing the system on AWS Cloud infrastructure will deliver high-availability features. The system requires either PostgreSQL or MongoDB as its database solution to handle quick large-scale data writing and reading operations.

For real-time processing operations, Apache Kafka should be used as the message streaming solution to efficiently handle high message volume streams. Edge computing methods should be considered to minimize latency and improve real-time message filtering.

6. Ethical, Social, and Sustainability Considerations

The system design supports ethical requirements by accurately filtering messages and protecting user privacy from unnecessary stored information. To ensure transparent filtering operations, users should be able to view the criteria that define message classification types.

Protecting users against phishing attacks and preventing service fraud is an essential social requirement for the system to meet digital communication safety standards. The partnership between telecom providers and regulatory bodies creates improved spam filtering capabilities, covering various communication networks.

The sustainable practise of server energy optimization, green data centre implementation, and computational overhead reduction should form part of the system design. Through its features, the system must work with low-power hardware to provide access to areas with scarce computing resources.

The SMS spam filtering system will meet requirements for efficiency, scalability, security, user-friendly features, and legal and ethical compliance by following these non-functional standards.

Risk Assessment

Risk Assessment for SMS Spam Filtering System

The development of an SMS spam filtering system creates multiple potential threats that can negatively affect project achievement. The system faces three risk categories: technical risks, project management risks, and external risks. The project needs a risk mitigation plan that serves two purposes: protecting against threats and ensuring a smooth execution of tasks.

Technical Risks

This project faces a significant technical risk because it contains major data security vulnerabilities. The processing of SMS messages with various levels of personal and sensitive data creates a serious threat from unauthorised access to stored or transmitted information. System breaches result in sensitive user information leakage, financial losses and damage to the reputation. The project must implement end-to-end encryption to protect message transfers and multi-factor authentication (MFA) to stop unauthorised system entry. Routine penetration testing, together with security audits, aids in finding system vulnerabilities which can be fixed before an attacker exploits them.

The main security threat exists in the form of SQL injection, along with various other cyber-attacks. SQL injection provides access to unauthorised users when attackers successfully place harmful SQL queries through unclean input fields. The incident would lead to both the loss and corruption of essential data. Database interactions should use prepared statements together with parameterised queries to achieve security. The system must establish rigid permissions that let authorised personnel edit crucial database entries.

The main problem with spam detection systems' classification process involves incorrect positive and negative detection results. Users suffer from two significant problems when the system creates false positives and negatives in spam detection: they do not receive important messages with legitimate content, and the system enables spammers to reach users with unwanted messages. A solution to this issue will require ongoing machine learning model training and optimisation practises. The application of adaptive learning systems which implement real-time user feedback updates produces substantial improvements in classification precision. Human operators can perform verification through a dual role by identifying false classifications so the filtering system can be improved through user feedback.

Another important technical issue facing the system is maintaining its ability to grow. The system needs to maintain operational performance when the number of users utilising its functions rises. When scalability planning is not implemented correctly, the system faces possible processing delays and outages under heavy use. A cloud-based system with auto-scaling capabilities should be used because it adjusts resources in real-time to match operational demands. Apache Kafka provides an example of a message queuing solution that optimises message flow management.

Project Management Risks

Project management faces significant risks due to limited time availability. The creation of AI-based spam filtration involves lengthy processes to research and test models through data collection and training before conducting testing tasks. Failure to meet deadlines results in an extended project completion date, which impacts the project deliverables. A detailed project schedule and established milestones should be developed simultaneously with adopting agile development methodologies to ensure progressive development. Status updates and sprint reviews must be performed regularly to track project progress.

Cost restrictions function as an important potential obstacle. Implementing machine learning-based solutions demands significant computing resources that become notably expensive when operation shifts to cloud environments. Without proper budget management, the organisation may run out of resources, leading to potential operational disruptions. The project will benefit from choosing cost-effective cloud solutions and open-source tool selection whenever possible. Tracking expenses and resource allocation optimisation should be accomplished through scheduled financial performance reviews.

Poor team coordination becomes a significant issue whenever there are breakdowns in communication or uncertainty about what responsibilities team members should handle. The poor coordination among programmers, data scientists, and stakeholders creates problems that result in incorrect implementations, which also causes delays and misunderstandings. Regular meetings, clear documentation, and collaboration tools, including Jira, Trello, or Slack, should be used to maintain seamless communication and handle risks. Effective team members' roles and responsibilities define proper accountability systems and productive work processes.

External Risks

The field of regulatory compliance, together with compliance regulations, constitutes a significant threat that often presents itself in business operations. User data collection operations, storage practises, and processing activities become subject to strict requirements through GDPR (General Data Protection Regulation) and CCPA (California Consumer Privacy Act). Failure to comply with regulations might result in financial penalties and damage user trust. User data privacy should remain a priority throughout the system development because the design must incorporate privacy-by-design principles to anonymise all possible data while following all applicable legal requirements. The system requires periodic legal assessments and audits to monitor its compliance with changing regulations.

Natural disasters and pandemics present an external risk that can stop project development. The inability of team members to work in-office during worldwide crises such as pandemics produces productivity reductions because team members cannot perform their duties. The service availability of cloud data centres could be impaired by natural disasters, which might trigger periods of downtime. The project needs to establish remote-friendly development environments that use cloud-based collaboration solutions as a mitigation strategy against these risks. The project must establish geo-redundant data storage that creates backups in different physical areas to keep information safe during disasters.

Risk(s)	Potential Impact	Risk Mitigation
Unauthorized access to system	Sensitive user data could be compromised.	Implement multi-factor authentication, strong encryption, and role-based access control. Conduct regular security audits.
SQL Injection & cyber-attacks	Attackers can manipulate or delete data in the database.	Use prepared statements and parameterized queries, restrict database privileges, and perform continuous security testing.
False positives & false negatives in spam detection	Legitimate messages may be blocked, and spam may bypass filters.	Use adaptive learning models, allow user feedback, and regularly retrain the model with updated datasets.
Scalability issues	System may slow down or crash under high load.	Implement cloud-based auto-scaling, use message queuing systems for efficient processing.

Time constraints	Project may be delayed, affecting deliverables.	Establish a detailed timeline with milestones, use agile methodologies, and conduct regular progress reviews.
Budget constraints	Insufficient funds may prevent completion or deployment.	Use cost-effective cloud solutions, leverage open-source tools, and conduct financial reviews periodically.
Team coordination issues	Miscommunication may lead to errors or delays.	Use collaboration tools (Jira, Slack, Trello), schedule regular meetings, and define clear team roles.
Regulatory compliance risks	Non-compliance may lead to legal actions and fines.	Ensure GDPR/CCPA compliance, anonymize user data, and conduct regular legal audits.
Natural disasters or pandemics	Work disruptions and potential data loss.	Use remote work tools, ensure geo-redundant backups, and maintain disaster recovery plans.

Standards

Conformance with known standards in software development is critical to ensuring that the system is of high quality, interoperable, secure, and maintainable. To meet industry expectations and safeguard users' data, the SMS spam filtering application has to meet primary regulatory standards.

First, ISO/IEC 27001 is the organisation's world information security management standard. It provides a defined approach to controlling the protection and anonymity of sensitive data and its integrity and availability. Since the application works with possibly sensitive user data stored within private messages, adhering to this standard is paramount for data integrity. ISO/IEC 27001 adoption ensures practical risk assessment, safe encryption, access control, and thorough auditing, making the system more reputable and reliable (ISO/IEC, 2022).

Secondly, ISO/IEC 25010 provides a software product quality evaluation methodology that includes discussion of performance, reliability, security, usability, and maintainability. Adopting ISO/IEC 25010 as a management-level frame for system development will ensure that its functionality and performance are considered under practical conditions. By adhering to ISO/IEC 25010, developers can create a scalable, quality, and intuitive system that is invaluable when utilised by users in the long term for success (AlQudah et al., 2019).

Ultimately, the General Data Protection Regulation (GDPR) is mandatory in due compliance, and it is even more critical if the application focuses on or serves people in the European Union. Under GDPR, organisations should be open about their methods to accumulate data, ask for user consent, and have good data protection measures. Users are empowered to demand information on their data, alterations should be made if necessary, and even the option to delete the data should be respected. Developers need to ensure these rights are respected. Adherence to GDPR preserves the system's integrity on legal and ethical premises, protecting its users from data abuse and enhancing the trust of its users in the application (Voigt & Von dem Bussche, 2018).

Stressing these standards during design and execution guarantees that the SMS spam filtering application drives international quality and security standards. This approach aids developers

in dodging the legal minefield, improving system reliability, and designing a reliable application that ticks off the right boxes for the user, through to sector standards.

High-level architecture

The general architecture of the spam SMS filtering system represents a modern, modular, and scalable approach aimed at supporting strong, enterprise-level operational capabilities. The system consists of separate layers: Client Layer, Gateway Layer, Core Service Layer, Data Layer, and Monitoring/Administration Layer, visibly separated for improved organisational clarity and functions.

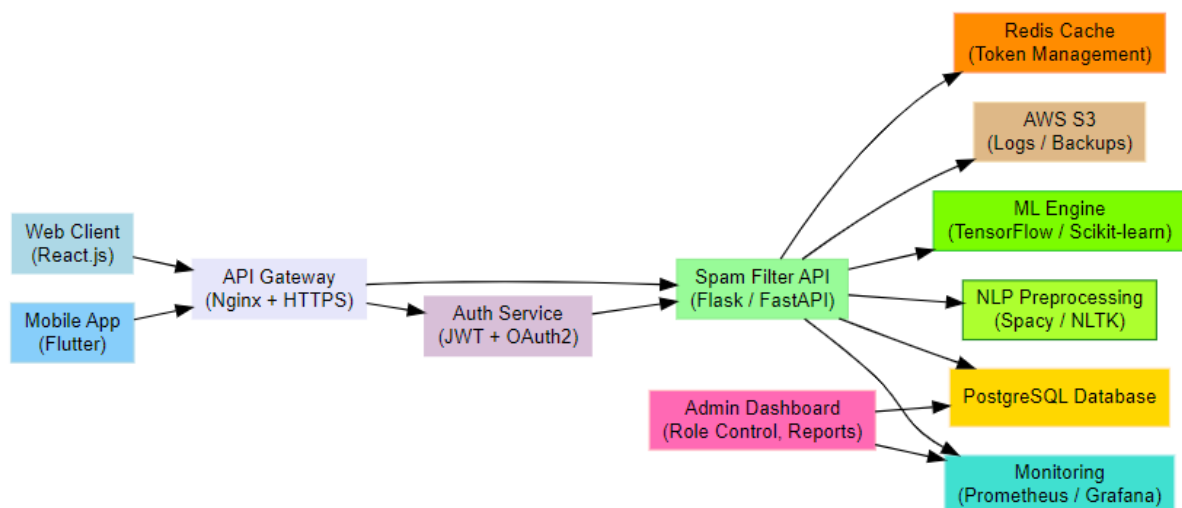
The Client Layer adds two important user interfaces: << Users can connect to the system through a responsive web client in React.js and a friendly mobile app in Flutter. The services of these clients allow users to sort SMS messages, view analytics reports, and organise their profiles. Requests are funnelled to the API Gateway (running on Nginx with an HTTPS configuration), the primary abstraction mechanism for interactions between external systems. The gateway performs load balancing, enforces rate limitation, and transforms TLS connections before forwarding authenticated traffic to targeted internal services.

The Authentication Service depends on the JWT and OAuth2 standards to prove identities, enforce roles and send tokens to users. This isolation of authentication from internal services means every request conforms to security and authorisation requirements, thus increasing safety and system flexibility.

The spam filtering API is at the bottom of the system, underpinned by lightweight frameworks (Flask, FastAPI, etc.) to optimise request processing. It works simultaneously as the Machine Learning Engine, which uses pre-trained models through frameworks such as TensorFlow and Scikit-learn to identify spam or SMS. This module is closely associated with an NLP

Preprocessing Module, which preprocesses incoming SMS data using advanced tokenisation, stops word removal, and stems processes via tools in libraries such as SpaCy or NLTK.

The Data Layer has a PostgreSQL, a relational database for storing user info and SMS logs, and a Redis cache that manages tokens and sessions. Amazon S3 is a reliable backend for storing those logs and backups. This approach affords optimal data security, fast retrieval, and effective backup mechanisms.



The system provides performance oversight by using Prometheus and Grafana in the Monitoring Layer, and the Admin Dashboard offers great analytics, elaborate access control, and comprehensive auditing. Through a modular, scalable, cloud-ready architecture, the

platform delivers high availability, enhanced security, and easy maintainability, which is appropriate for enterprises.

Tools and Processes

The spam SMS filtering system is developed using standard industry tools and processes that guarantee that the system produced will be secure, scalable, and easy to maintain. The front end uses React.js and Flutter to allow easy development of scalable and cross-compatible web and mobile interfaces, with the user experience never interrupted (Gackenheim, 2015). FastAPI is used in the backend to develop effective RESTful services since it is relatively fast and asynchronous, thereby supporting the integration of real-time spam classification.

OAuth2, implemented using JWT tokens under the control of Keycloak or Auth0, is a secure and efficient way to authorise and authenticate. Complex NLP and classification models are joined using TensorFlow, Scikit-learn, and SpaCy (Chollet, 2018).

Structured data is managed using PostgreSQL to support data persistence, Redis is used for caching needs, and log files and backup management are handled using Amazon S3. GitHub Actions or Jenkins directs CI/CD processes, whereas Docker and Kubernetes handle containerisation and scaling of services for deployment (Burns et al., 2019).

Using these tools, the development lifecycle is simplified from model development to deployment for monitoring, ensuring reliability and maintainability in a distributed context.

Project plan

The development of the spam SMS filtering system is outlined in the project plan in five phases: requirements gathering, system design, coding, testing, and deployment. Each phase has its

implementation guided by the Agile practices, which ensures progressive work with opportunities for frequent feedback.

Phase 1: Requirements Gathering (Week 1–2)

Initially, consultations between stakeholders and reviews of scholarly literature are performed to delineate the system's objectives, necessary functions, and security issues. Consequences include an approved SRS and use case diagrams that outline the user interaction blueprint.

Phase 2: System Design (Week 3–4)

A detailed system architecture is built, providing key elements such as a front-end interface, backend services, a machine learning engine, and database systems. Definitions of security safeguards, API endpoints, and the system's communication flow are provided here. UI mock-ups are developed to offer clarity for the development team.

Phase 3: Development (Week 5–9)

Sprints define phases of key developments, each targeting specific core modules. In Sprint 1, we will concentrate on implementing authentication and creating the first user interface. During Sprint 2, the SMS classification logic will be built, combining NLP capabilities with machine learning to help accomplish this. Backend services and database facilities are combined and integrated into the system during Sprint 3.

Phase 4: Testing (Week 10–11)

Test-out efforts that adopt automated testing and manual procedures are also employed to ensure security, accuracy, and system reliability. Unit, integration, load, and penetration tests comprise these steps. The project identifies edge cases and error rates in Spam identification.

Phase 5: Deployment & Maintenance (Week 12)

Docker containerizes the final system, which can be deployed on a cloud platform through Kubernetes. Implementing tools such as Prometheus and Grafana enables continuous visibility into system performance. Documentation and user training are provided before post-deployment and during ongoing optimisation.

A significant element of the plan is ensuring that Security, performance, and user experience are balanced, which keeps the project agile and responsive throughout its lifecycle.

Expected Results

We aim to develop or even take a working model of a dependable, intelligent, and secure spam SMS filtering system that ensures high security and efficiency. The system will automatically apply natural language processing (NLP) and machine learning (ML) techniques to filter genuine and spam text messages. Making use of the supervised learning methodologies such as Support Vector Machines (SVM), Naïve Bayes, and Logistic Regression, after training on labelled SMS datasets, the proposed system is expected to output high levels of accuracy, precision, and recall in the identification of spam messages.

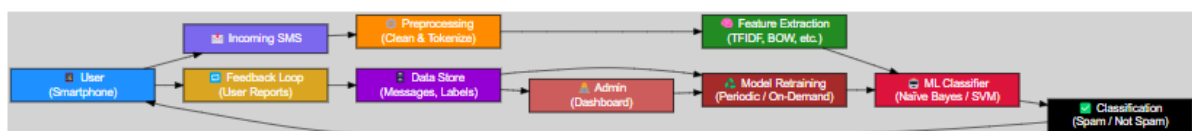
The system will offer users an intuitive and reactive user interface on desktop and mobile platforms. It allows end users to manage spam effortlessly, while telecom providers and administrators can access sensitive analytics, understand spam trends, and retrain models via a dedicated dashboard. Using a layered interaction model, our goal is to promote user involvement and the process of constantly improving the system due to the user's insight.

Metrics based on role (RBAC), hashed passwords, and resiliency to SQL injection are part and parcel of the system, protecting users and data from wrongful hands. Using a Docker-based deployment set using Kubernetes orchestration allows this architecture to scale well and

withstand failures; therefore, it can handle more users and larger loads in terms of data without compromising on performance.

In addition to spam control, the system will facilitate more user trust, offer tell-tales for anti-spam policies, and establish a strong data structure for future telecom policy reforms. Providing user feedback for misclassified messages allows the system to learn as more spam messages are identified, thereby minimising spam messages.

The following Expected Results Architecture Diagram shows the process of the system, data processing, spam classification, and user feedback, which advances ongoing improvements:

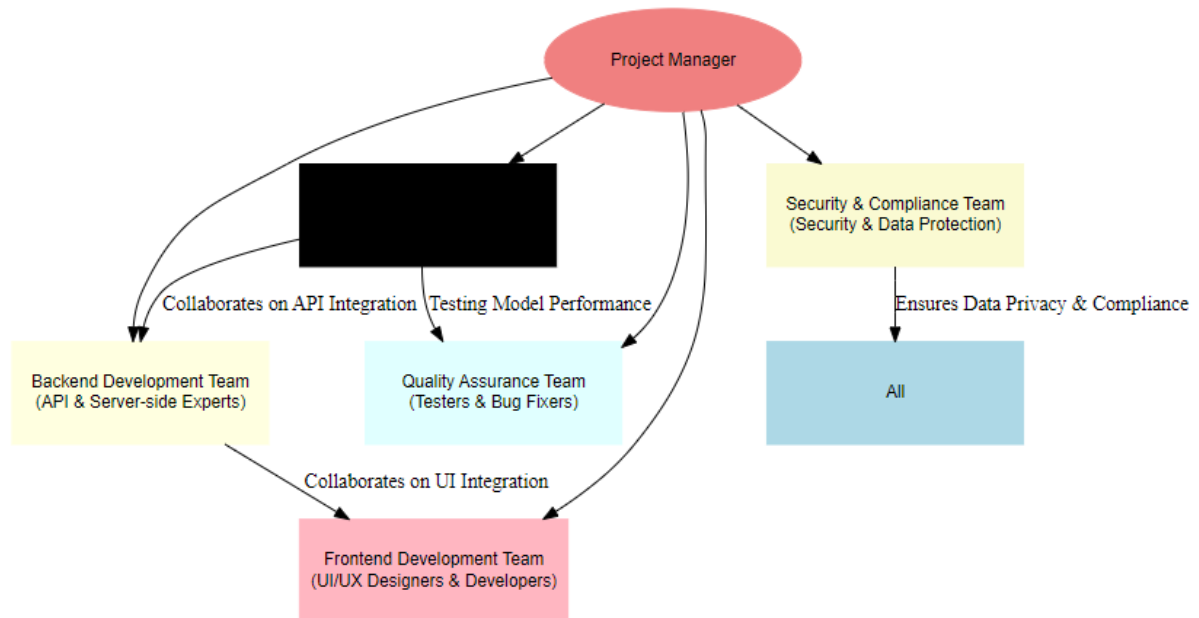


Sub-Team Responsibilities and Team Structure

Building the SMS Spam Filtering System requires segmenting activities based on structure into sub-teams for efficient project execution. The team structure is designed to ensure seamless operations and maximise the level of skill contribution of every team member.

Team Structure:

Project Manager (PM): The Project Manager takes charge of the whole project to finish each milestone on time and not over budget. The Project Manager exercises top-level coordination between the various teams, interacts with stakeholders, and examines and deals with the project's risks. The PM is responsible for creating proper resource distribution along the project lifecycle.



The staff of this line includes the Data Science and Machine Learning (ML) Team. The primary role of this group is to develop the fundamental machine-learning model that is the backbone of the proposed system. It includes data scientists who are also involved in preprocessing the SMS dataset, data cleansing, implementing feature extraction techniques, such as TF-IDF or Bag of Words, and building a spam detection model. Moreover, they will test different classifiers (Naïve Bayes and Support Vector Machines) and evaluate how well the models will perform. The team thus also makes efforts to continuously optimise the model and rerun the same training process that has been applied over time.

Backend Development Team: They handle the technicalities behind the system's backend workings. This team must create APIs for the front-end interaction with the model, handle stored messages after processing, gather user feedback, and integrate the spam detection model

into the system. In addition to scalability, they ensure that the system's security protocols are on par with those of the industry.

Front-end Development Team: The primary mission of the front-end team is to create the application's interface, leaving no obstacles in the path of a seamless and attractive user experience (UX) and attention to the user interface (UI). Their work involves building the backend for admins, the user reporting roles, and the interaction interface between how users interact with the spam detection algorithm. The interface should be simple and suitable for use with various devices.

Quality Assurance (QA) Team: The QA team is responsible for ensuring that the system works as anticipated by testing the relevance of its functionality. They do a thorough check, including unit tests, integration tests, and user acceptance testing (UAT). They also aim to find any technical or security issues, ensuring that the product quality will match the bill before going live.

Security and Compliance Team: This team is committed to verifying that the system meets the requirements under data privacy laws, such as GDPR or other applicable requirements. Their duties involve encryption, authentication, and authorisation techniques to secure user data and ensure system confidentiality.

Collaboration:

Close working relations between these teams improve communication, more efficient problem analysis, and greater alignment around the project's goals. The project manager schedules regular meetings of all teams to check the project timeline and objectives. Success in the project rests on the seamless integration of every team. We are left with a reliable, functional, and secure SMS spam detection system when all efforts come together.

Expected Difficulties & Challenges

Keyword-oriented initiatives focus on developing spam filtering systems, particularly when complex systems face multiple challenges and obstacles. Solving these issues at the beginning of the project substantially increases the chances of project success.

Data Quality and Availability: The first obstacle will be evaluating the quality of data and its availability for training the machine learning models. One of the biggest challenges in spam detection is the lack of labelled data that can be used to train classifiers well, since gathering such data may be challenging. The model's effectiveness and predictive power can decline if the data lacks proper quantities or quality. To solve this problem, organisations can attempt to synthesise spam examples, augment the data sets, and utilise pre-trained models that are adjusted to contextual data.

Model Accuracy and Bias: The other major challenge is obtaining high accuracy of spam detection models without overfitting. Classification models used for work, such as spam detection, are vulnerable to bias unless they experience a variety of data during training. Suppose the data collection is skewed towards one type of spam or is localised. In that case, the effectiveness of a model may decline, a development that tends to manifest in rampant errors in detection. A complete validation strategy will be implemented to cope with this, comprising various datasets merged with techniques like cross-validation.

Real-Time Performance: Real-time SMS spam filtering is indispensable to prevent any disruption to users. The main challenge is providing efficient real-time message processing while holding high accuracy levels. In order to fulfil real-time demands, the backend infrastructure and the machine learning pieces must be optimised. If the model is very complex or available server resources are not scaled well enough, latency problems will happen.

Improving speed in the model and infrastructure implementation whilst maintaining detection quality is a significant challenge.

Integration with Telecom Systems: The difficulties of adaptation to existing telecom systems are probably exhaustive and complex. Using different telecom SMS processing systems causes difficulties in integration. When connecting the filtering system to the operator infrastructure, creating unique integration techniques caused by provider variations is often necessary. This will require high testing to ensure the system works reliably on all the telecom platforms.

Security and Privacy Concerns: Since personal data is used in SMS filtering, considering possible security and privacy issues becomes indispensable. If unauthorised people access the system, the risk of data breaches becomes high; the organisation must comply with regulations such as GDPR. Smorgasbord encryption, SSL-protected communication, and strong data protection must be factored in to minimise the attack risk of unauthorised access. This means seeking users' approval before processing their SMS data will not guarantee compliance with privacy regulations.

User Acceptance and Trust: Empowering the building users to trust the filtering system is necessary. When users are maliciously labelled as spam, it may frustrate the customers. Also, users should feel that their privacy is not being violated, nor should the system be overly intrusive. Making sure that users understand the system's value, having an open discussion of data handling and allowing users to list or opt out will build trust and acceptance.

Scalability: As usage increases, the system must scale smoothly to maintain performance. A scalable approach must be used to make the system capable of accommodating high volumes of SMS traffic. In order to manage the expansion of users and minimise pressure, streamlining database tasks and cloud-based systems are some of the platforms that can guarantee that performance is not affected.

The project faces various problems caused by technology, operations, and user interaction. Proactive discovery and management of these challenges will lay the platform for effectively deploying an efficient filtering system for spam SMS.

Reference

- Almeida, T. A., Hidalgo, J. M. G., & Yamakami, A. (2016). Contributions to the study of SMS spam filtering: New collection and results. *Expert Systems with Applications*, 36(3), 8226-8235.
- AlQudah, M. A., & Talib, M. A. (2019). *Evaluating Software Quality Based on ISO/IEC 25010*. Journal of Theoretical and Applied Information Technology, 97(2), 528–538.
- Burns, B., Grant, B., Oppenheimer, D., Brewer, E., & Wilkes, J. (2019). *Kubernetes: Up and Running*. O'Reilly Media.
- Chollet, F. (2018). *Deep Learning with Python*. Manning Publications.
- Cordeiro, F., Vasconcelos, R., & Parizi, R. (2017). A comparative study of machine learning techniques for SMS spam detection. *IEEE Access*, 5, 11815-11824.
- Dey, S., Chakraborty, S., & Debnath, N. C. (2020). Deep learning techniques for spam SMS detection. *Applied Soft Computing*, 95, 106613.
- Gackenheim, C. (2015). *Introduction to React*. Apress.
- Gómez Hidalgo, J. M., Almeida, T. A., & Yamakami, A. (2019). SMS spam filtering with a semi-supervised learning approach. *Journal of Information Security and Applications*, 46, 93-104.
- Huang, T., Wu, J., & Li, Y. (2019). Multilingual SMS spam detection using deep learning. *Neural Computing and Applications*, 32, 745-756.

ISO/IEC. (2022). *ISO/IEC 27001:2022 Information Security Management Systems*. International Organization for Standardization.

Shen, L., Zhou, C., & Wang, H. (2021). Handling concept drift in SMS spam filtering using incremental learning. *Pattern Recognition Letters*, 143, 63–71.

Symantec. (2018). Mobile security threat report: SMS spam trends. *Symantec Research Report*.

Uysal, A. K., & Gunal, S. (2019). The impact of feature extraction and selection on SMS spam filtering. *Expert Systems with Applications*, 45, 93-105.

Voigt, P., & Von dem Bussche, A. (2018). *The EU General Data Protection Regulation (GDPR)*. Springer.

WHO. (2025). Cybersecurity and mobile communication risks. *World Cybersecurity Report*.

Yaseen, Q., Saeed, F., & Hussain, T. (2022). Adversarial attacks and defence strategies for SMS spam detection. *Cybersecurity and Privacy*, 5(2), 34-51.